ALOK SINGH KATARIA

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PHYSICS · COMPUTATIONAL SCIENCE · SOFTWARE ENGINEERING · AI · QUANTUM COMPUTING

OBJECTIVE

Seeking a role in **academic research, EdTech, or academic software development** where I can combine my foundation in **Physics, Mathematics, and Numerical Methods** with hands-on software engineering and DevOps experience. Specifically interested in opportunities involving **computational science, scientific software, educational platforms, or research tooling** — where rigorous mathematical thinking, numerical simulation, and modern software practices intersect to solve meaningful problems in academia or education.

PROFILE

Physicist and Senior Software Engineer with 25+ years of professional experience across software development, cloud platforms, DevOps, automation, enterprise systems, and EdTech. **IIT Kanpur Physics graduate** with a strong foundation in quantum mechanics, mathematical physics, linear algebra, probability, and numerical methods.

Hands-on experience with Python, scientific computing, AI/ML, and quantum-computing tools. Independent technical work includes **quantum mechanics, numerical methods, Hamiltonian and eigenvalue problems, VQE, quantum simulation, quantum chemistry, and quantum optimization** using Python and Qiskit.

Combines Physics and mathematical foundations with extensive software-engineering experience to **model, simulate, analyze, and prototype computational solutions**. Also experienced in developing mathematical and educational software and working with AI/ML approaches in EdTech.

Interested in opportunities involving Physics, computational science, scientific software, EdTech, AI/ML, and practical applications of quantum computing.

CORE AREAS

`Computational Physics` · `Quantum Mechanics` · `Mathematical Physics`
`Scientific Computing` · `Numerical Methods` · `Linear Algebra` · `Optimization`
`Quantum Computing` · `Quantum Algorithms` · `Quantum Simulation`

VQE · Quantum Chemistry · Quantum Optimization
Quantum Noise & Error Mitigation

SELECTED QUANTUM & COMPUTATIONAL PHYSICS WORK

- Applied classical numerical methods to quantum systems including the **hydrogen atom, 1-D particle-in-a-box, and harmonic oscillator**, using Python, NumPy, SciPy, linear algebra, and the **Finite Difference Method (FDM)** to construct Hamiltonians and compute energy eigenvalues, comparing numerical results with analytical solutions where available.
- For systems without convenient closed-form solutions, explored approximate numerical approaches using Python and **AI-assisted computational methods** for formulation, implementation, and validation.
- Applied a similar computational approach to quantum-computing formulations, using **qubit representations, Hamiltonians, quantum circuits, and VQE** with Python and Qiskit.
- Explored quantum-chemistry workflows using **STO-3G, Hartree–Fock, molecular Hamiltonians, and variational ansätze** for ground-state energy estimation, comparing results with classical calculations.
- Used classical numerical solutions as reference points for exploring corresponding **quantum formulations and variational approaches**, along with quantum optimization and quantum-information concepts such as portfolio optimization, qubit encoding, quantum randomness, spin dynamics, Hadamard transformations, and measurement probabilities.

PROFESSIONAL EXPERIENCE

IBM — Advisory Software Engineer | DevOps / SRE | 2022 – Present

1. CSUtils Ecosystem — Platform Architecture, Resilience & L3 Engineering

- Led architecture modernization and resilience engineering across IBM Cloud security utilities, modernizing Kubernetes configuration with CRDs, strengthening VPN connectivity with fail-fast validation, and ensuring reliable syslog delivery across pod restarts.
- Designed automated multi-cluster chaos testing and owned L3 production troubleshooting across Kubernetes, VPN, TCP/UDP connectivity, and networking failures to improve service reliability.

2. Dreadnought — IBM Cloud PaaS Re-platforming & Cloud Engineering

- Drove IaC-based provisioning and re-platforming of IBM Cloud infrastructure, including VPC networking, OpenShift, IAM, managed databases, and CI/CD integrations using standardized platform configurations.
- Implemented private endpoint connectivity, PAG/GPVPN integration, bastion access, and security scanning; supported complex cloud networking and platform escalations through root-cause analysis and remediation.

3. Compliance Automation — AI Agent, Auditree & CI/CD Engineering

- Led Auditree onboarding and Python-based compliance automation for SOC 2-aligned evidence collection, security posture assessment, centralized evidence storage in IBM Cloud COS, and Git-based remediation workflows.
- Initiated an AI agent for troubleshooting Jenkins and Auditree pipelines, leveraging repository documentation for failure analysis and remediation recommendations; integrated automated workflows with CI/CD practices spanning Jenkins and Tekton.

4. Armada Compliance Engine — Compliance Automation, SRE & DR Hardening

- **Led Compliance Modernization & Process Improvement:** Spearheaded the transition from legacy, high-latency node-level BigFix agents to a centralized, automated Go-based event-driven compliance engine, drastically reducing inventory lag and automating real-time registration/deregistration at scale.
- **Headed Incident Resolution & Diagnostic Optimization:** Resolved critical, silent compliance-sync defects (SOS API silent failures); designed and published an end-to-end operational runbook with standard decision trees, search signatures, and `@xo` CLI diagnostic flows, reducing mean-time-to-resolution (MTTR) for on-call teams.
- **Engineered DR Resilience & CI/CD Integrity:** Enforced disaster recovery (DR) readiness by configuring zone-spread constraints and high scheduling priorities (`system-cluster-critical`) to prevent node-starvation outages during brownfield migrations, while debugging and resolving broken Jenkins regional carrier lookup and build env pipelines.

Wipro — Lead Software Developer | 2022

Software development for the **CreditMemo** process of the Commissioning System for **Telefónica Germany**, building Python/Flask/Pandas services for data loading, aggregation, and validation.

- Implemented feature enhancements, defect fixes, and unit testing; worked with PostgreSQL and Google Cloud.

RevenueBox — Senior Software Engineer | 2021

B2B Account-Based Marketing (ABM) and go-to-market software focused on identifying, prioritizing, and engaging the right customer accounts.

- Developed a Python microservice to identify, process, and select relevant customer accounts based on multiple input signals.
- Designed a mathematical scoring model to calculate and assign a "**Quibble Score**" for account prioritization and tracking, using Python, Docker, MongoDB, and Google Cloud.

Lumos Learning — Academician / Educator & Research Scientist | 2019 – 2020

EdTech platform delivering dynamic, animated, and guided-practice learning experiences for **K-12 students**, aligned to Common Core Standards (corestandards.org), with emphasis on mathematics, automated question generation, and illustrative scientific concepts.

Drove the platform's modernization end to end — uniquely bringing **Physics, Mathematics, animation-driven software development, and serverless cloud architecture** together in one frame

— and led the addition of several new modules spanning dynamic question creation, guided practice, and a mathematics chatbot.

- Led development of **Python-based animated and interactive question-generation modules** that expose K-12 students to dynamic, visually engaging problems with **varied difficulty levels** — catering to the full ability spectrum, from remedial to advanced.
- Architected and built **guided-practice workflows** with automated solution processing, difficulty estimation, and adaptive evaluation, enabling step-by-step problem solving aligned to Common Core standards.
- Led the design of a **mathematics-focused chatbot** and educational automation initiatives, applying NLTK-based natural language processing and classical neural-network/machine-learning techniques of the period.
- Designed the **serverless cloud architecture** using Python, AWS Lambda, and SQL Server to support these modules.
- Illustrated advanced mathematical and scientific concepts — including **solving quantum Hamiltonians via the Finite Difference Method and Fourier Transform techniques** — in simplified, visual, and accessible ways suitable for curious students and educators.

HCL Technologies — Senior Software Engineer | 2017 – 2018

L3 product support for **IBM AppScan** (web application security testing), troubleshooting complex customer issues, root-cause analysis, and coordinating fixes across SAST/DAST scanning and reporting workflows.

- Worked on AppScan SAST/DAST capabilities, customer configurations, defect analysis, and production issue resolution.

IBM India Software Labs — Senior Staff Software Engineer | 2008 – 2017

L3 Development & Troubleshooting Engineer for **IBM InfoSphere Data Replication (IIDR)** (previously DataMirror), specializing in complex production issues, root-cause analysis, debugging, and customer escalations.

- Expert in heterogeneous data replication and integration, including **DB2 for i / AS400, Oracle, DB2, SQL Server, and Teradata**, with IBM messaging and remote journaling technologies.
- Strong experience in cross-language and cross-platform integration, including **AS400/IBM i** and COBOL-based enterprise systems; supported customer ETL and data-movement workflows.
- Recognized as a performance specialist, contributing to performance analysis and optimization and receiving an **IBM award for performance expertise**.
- Contributed to feature development, product maintenance, automation, and build/release pipeline engineering across multiple product releases.
- Worked with C, Java, Python, RPG, COBOL, AS400, and Jenkins.

CA Technologies — Software Engineer | 2005 – 2008

Worked on **eTrust Access Control**, **eTrust Audit**, and **eTrust Single Sign-On** — enterprise security products.

- Developed more than 100 automated tests for nightly builds and ported product code to AIX.
- Investigated customer and QA defects including memory corruption, enhancements, and installation/upgrade issues.
- Worked with C, Python, Linux, UNIX, AIX, Solaris, and HP-UX.

TECHNICAL SKILLS

Programming: Python, C, Java, Shell, JavaScript

Scientific Computing: NumPy, SciPy, Pandas, MATLAB, scikit-learn, TensorFlow

Quantum Computing: Qiskit, VQE, Quantum Circuits, Quantum Simulation

AI-Assisted Engineering: ChatGPT, IBM Bob, GitHub Copilot (leveraged for codebase analysis, debugging, refactoring, and structured document engineering)

Cloud & DevOps: IBM Cloud, AWS, Google Cloud, Kubernetes, OpenShift, Docker, Jenkins, Tekton, CI/CD

Databases: PostgreSQL, MongoDB, SQL Server, Oracle, DB2, Teradata

Operating Systems: Linux, AIX, Solaris, HP-UX, Windows, IBM i / AS400

PERSONAL PROJECTS | RESEARCH & ACADEMIC

Independent projects undertaken for research, academic exploration, and applied learning — combining quantum computing, AI, DevOps, and scientific computing.

Book Knowledge Service

`quantum-computing / book-knowledge-service` github.com/alokkataria101/quantum-computing

FastAPI microservice that transforms physics PDF textbooks into a searchable, AI-powered knowledge base for study and research.

- PDF ingestion, chapter detection, SQLite FTS5 full-text search; pattern-based question extractor for Examples, Exercises, Objective I/II, Short Answer sections with page tracking
- Browser UI with page reader, question browser, figure thumbnails, and IBM watsonx (llama-3-3-70b) RAG Q&A and AI question generation
- Containerised with Docker + docker-compose; shared secrets architecture; zero-restart Jinja2 template updates

Quantum Portfolio Intelligence Platform (QPIP)

`quantum-computing / stock-analyzer` github.com/alokkataria101/quantum-computing

Streamlit investment dashboard applying quantum-inspired signal processing and IBM watsonx AI for NSE portfolio analysis.

- Quantum state similarity matching generates BUY/ADD/HOLD/REDUCE/AVOID signals with probability distributions
- IBM watsonx (Granite/llama) produces natural-language signal explanations and analyst-style research reports
- Portfolio Doctor, What-if Simulator, Opportunity Scanner, live intraday feed, adaptive memory; containerised with Docker

Quantum Computing Research Suite

`quantum-computing / Hamiltonian · VQE · QFT · phase-estimation`
github.com/alokkataria101/quantum-computing

End-to-end quantum computing experiments from first principles through Qiskit circuit implementations.

- FDM-based Schrödinger solvers for particle-in-a-box and hydrogen atom; Hartree–Fock vs VQE comparison
- Five-stage VQE: HF → helium → H_2 dissociation → CO_2 active space → CCSD benchmark on Qiskit Aer
- QFT applied to NSE financial time series and mathematical functions — QFT vs classical FFT comparison
- Full Quantum Phase Estimation, Grover's algorithm, and Shor's algorithm demonstrations

SELECTED RESEARCH & TECHNICAL WORK

Quantum Intuition: From Spin Dynamics to the Hadamard Gate and π Estimation via Quantum Randomness

Explored quantum randomness and quantum-state transformations through Qiskit-based simulations, including spin dynamics, Hadamard transformations, measurement probabilities, and Monte Carlo estimation of π.

Status: Under IBM internal review for publication.

Classical VQE Demonstration for the Hydrogen Atom

Developed a classical demonstration of the VQE approach for the hydrogen atom, connecting the variational principle, Hamiltonian formulation, parameterized states, energy expectation values, and numerical optimization.

Status: Under IBM internal review for publication.

INTELLECTUAL PROPERTY

- **US Patent #9,528,562** — Granted US patent related to mechanics.
- Two additional mechanics-related inventions under IBM patent review.

EDUCATION

IIT Kanpur — Integrated M.Sc. in Physics | 2000

Academic foundation includes Quantum Mechanics, Classical Mechanics, Electromagnetism, Mathematical Physics, Linear Algebra, Probability, Differential Equations, Numerical Methods, and Statistical Physics.

ACADEMIC DISTINCTIONS & AWARDS

- **All India IIT-JEE Advanced Rank** — 1211 (General)
- **GATE All India Rank** — 29 (General) | **Score** — 98.73 (General)
- **Roorkee Entrance Exam 1994** — Rank 1741 (General)
- **Bihar Engineering Entrance Test** — Rank 452 (General)
- **IBM Excellence and Eminence Award (2012)** — Received for outstanding L3 support, troubleshooting expertise, and high-priority resolution of complex client issues.

PHYSICS TEACHING & MENTORING | IIT-JEE ADVANCED / NEET FACULTY

Institutions: Vizag Vikas College · Narayana Junior College, Kurnool · SRM · PIE Education

- **Vizag Vikas College (2001–2004):** Contributed to 40+ IIT-JEE selections including AIR 155; designed original IIT-JEE Advanced-level questions; supported competitive examinations and faculty recruitment across Andhra Pradesh.
- **Narayana Junior College, Kurnool (2017–2018):** Contributed to IIT-JEE and AIEEE selections; reviewed and refined questions from Narayana's curriculum and textbooks for accuracy, originality, and difficulty calibration.
- **NEET & CBSE Physics — SRM:** Taught Physics for medical entrance and CBSE curricula, emphasizing conceptual understanding and systematic problem-solving.
- **Teaching Methodology:** First-principles derivations, mathematical formulation, conceptual depth, and rigorous problem-solving from fundamentals to IIT-JEE Advanced level.